

Phishing Email Analysis and IOC Extraction Report

Project: Phishing Email Analysis and IOC Extraction Workflow

Case Study: Cancel Your Order

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Analyst: Anji Gajula

Classification: Phishing Email

1. Executive Summary

This report analyzes a phishing email that impersonates PayPal by sending a fake payment receipt. The attacker attempts to create urgency by informing the recipient of an unauthorized purchase and encouraging them to cancel the transaction. The email uses sender spoofing, URL shortening, and branded HTML content to appear legitimate and redirect users to a malicious website.

2. Scenario Overview

The phishing email claims that a payment has been made for an online purchase and provides a "Cancel the Order" button for users who did not authorize the transaction. The objective of the attacker is to trick the recipient into clicking the embedded link, which may lead to credential theft or financial fraud.

3. Phishing Techniques Identified

Spoofed Sender Identity

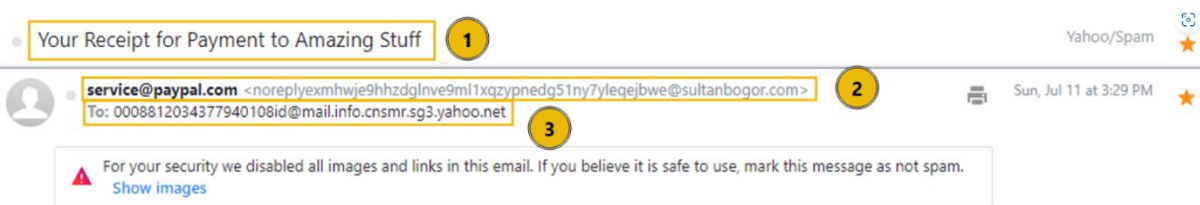
- The email appears to be sent from **service@paypal.com**.
- The actual sender address (**gibberish@sultanbogor.com**) does not belong to PayPal.
- This mismatch is a strong indicator of sender spoofing.

URL Shortening

- The attacker hides the actual destination using a shortened or redirected URL.
- Users cannot easily verify where the button will redirect them.

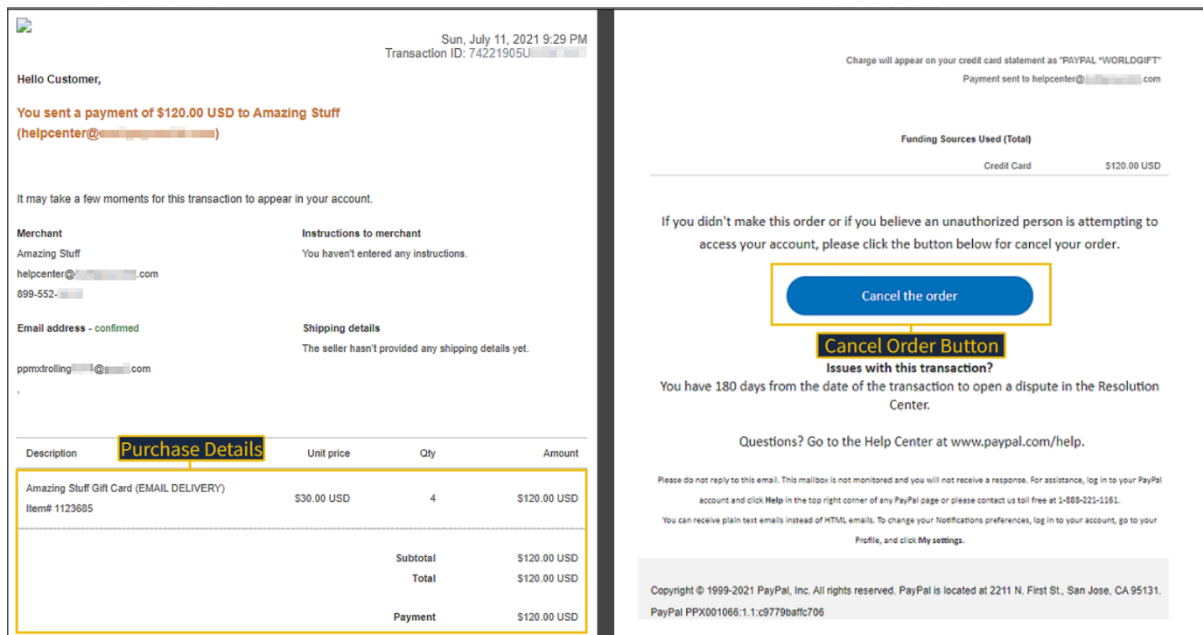
Branded HTML

- The email uses PayPal branding, logos, and formatting.
- This creates a false sense of trust and authenticity.



4. Indicators of Compromise (IOCs)

IOC Type	Value
Display Name	service@paypal.com
Actual Sender Domain	sultanbogor.com
Brand Used	PayPal
Subject Theme	Payment Receipt
Attack Category	Phishing
Delivery Method	Malicious Hyperlink
Action Requested	Cancel the Order



5. Analysis Findings

Subject Analysis

The subject line claims that a payment has been processed, creating urgency and encouraging the recipient to react immediately.

Sender Analysis

Although the display name appears to be PayPal, the actual sender domain does not belong to PayPal, indicating email spoofing.

Content Analysis

The email uses official PayPal branding and claims that the recipient has purchased gift cards. The "Cancel the Order" button is intended to persuade users to click before verifying the transaction.

Hyperlink Analysis

The "Cancel the Order" button is the primary attack vector. Clicking it may redirect victims to:

- Fake PayPal login pages
- Credential harvesting websites
- Financial phishing pages
- Malware download websites

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<!-- CTA text 17/21 -->
<td bgcolor="#0070BA" align="center" valign="middle" style="padding:13px
90px 13px 90px; display: block; color:#ffffff !important; text-decoration:none; white-space: nowrap;
-webkit-print-color-adjust: exact; display: block; font-family:Calibri, Trebuchet, Arial, sans serif;
font-size:17px; line-height:21px; color:#ffffff !important;border-radius:25px;" class="ppsans
mobilePadding9"><a type="Link" isLinkToBeTracked="true" target="_blank"
linkId="e854f9fff3d47fb841257f9bebb4d1b6" style="text-decoration:none;color:#ffffff !important;
white-space: nowrap;" href="https://is.gd/6oCJ4m?#-u7g?ah7Je1" Cancel the order</a></td>
<!-- end CTA text --></tr>
```

<https://is.gd/6oCJ4m>

Expand

Screenshot of the distant page behind your short URL



6. Detection Opportunities

- Inspect embedded URLs.
- Detect URL redirection or shortened links.
- Block known phishing domains.

7. Risk Assessment

Risk Level: High

Severity: High

Confidence: High

Category: Phishing / Credential Theft / Financial Fraud

Reasons

- Trusted PayPal brand impersonation
- Sender email mismatch
- Hidden malicious hyperlink
- Urgency-based social engineering
- Potential credential and financial theft

8. MITRE ATT&CK Mapping

Technique ID : T1566.002

Technique Name : Phishing: Spearphishing Link

The attacker delivers a phishing email containing a malicious hyperlink that redirects the victim to a fraudulent website for credential theft, malware delivery, or other malicious activity.

9. Recommended Actions

1. Do not click the "Cancel the Order" button.
2. Verify recent transactions directly through the official PayPal website or mobile application.
3. Block the sender's email address and domain.
4. Report the email as phishing.
5. Inspect embedded URLs using security tools before accessing them.
6. Enable Multi-Factor Authentication (MFA) on financial accounts.

10. Conclusion

This email is a phishing attempt that impersonates PayPal to trick recipients into clicking a malicious link by claiming that an unauthorized purchase has been made. Multiple indicators—including sender spoofing, branded HTML, URL manipulation, and urgency-based social engineering—confirm that the email should be classified as malicious and blocked.